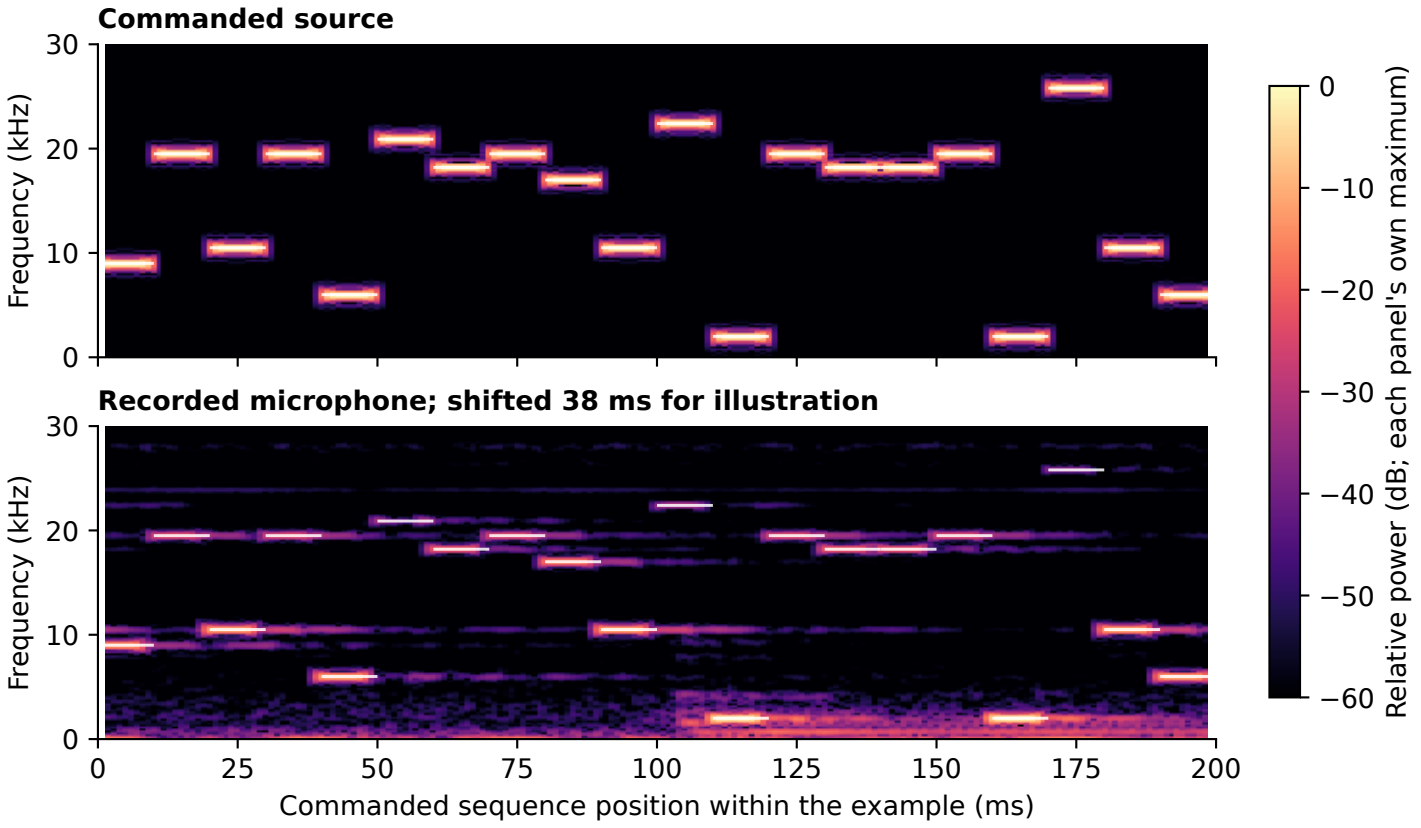


White segments: commanded frequencies. Microphone display alignment is illustrative; no SONIC-specific delay is inferred.
4 ms Hann window; 1 ms hop; ≈250 Hz window scale (zero-padded FFT). This view does not qualify individual 10 ms pip envelopes.